CD40110B Types

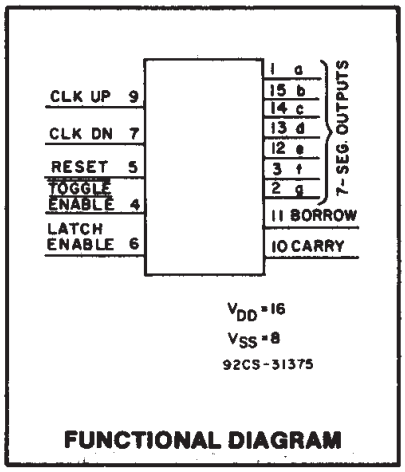
CMOS Decade Up-Down Counter/Latch/Display Driver

High-Voltage Type (20-V Rating)



Features:

- *Separate clock-up and clock-down lines*
- *Capable of driving common cathode LEDs and other displays directly*
- *Allows cascading without any external circuitry*
- *100% tested for quiescent current at 20 V*
- *Maximum input current of 1 μA at 18 V over full package-temperature range; 100 nA at 18 V and 25° C*



FUNCTIONAL DIAGRAM

■ CD40110B is a dual-clocked up/down counter with a special preconditioning circuit that allows the counter to be clocked, via positive going inputs, up or down regardless of the state or timing (within 100 ns typ.) of the other clock line.

The clock signal is fed into the control logic and Johnson counter after it is preconditioned. The outputs of the Johnson counter (which include anti-lock gating to avoid being locked at an illegal state) are fed into a latch. This data can be fed directly to the decoder through the latch or can be strobed to hold a particular count while the Johnson counter continues to be clocked. The decoder feeds a seven-segment bipolar output driver which can source up to 25 mA to drive LEDs and other displays such as low-voltage fluorescent and incandescent lamps.

A short durating negative-going pulse appears on the BORROW output when the count changes from 0 to 9 or the CARRY output when the count changes from 9 to 0. At the other times the BORROW and CARRY outputs are a logic 1.

The CARRY and BORROW outputs can be tied directly to the clock-up and clock-down lines respectively of another CD40110B for easy cascading of several counters.

- *Noise margin (full package-temperature range) =*
  1 V at V_{DD} = 5 V
  2 V at V_{DD} = 10 V
  2.5 V at V_{DD} = 15 V
- *5 V, 10 V and 15 V parametric ratings*
- *Meets all requirements of JEDEC Tentative Standard No. 13B, "Standard Specifications for Description of 'B' Series CMOS Devices".*

Applications:

- *Rate comparators*
- *General counting applications where display is desired*
- *Up-down counting applications where input pulses are random in nature*

The CD40110B types are supplied in 16-lead dual-in-line ceramic packages (D and F suffixes), and 16-lead dual-in-line plastic package (E suffix), and also available in chip form, (H suffix).

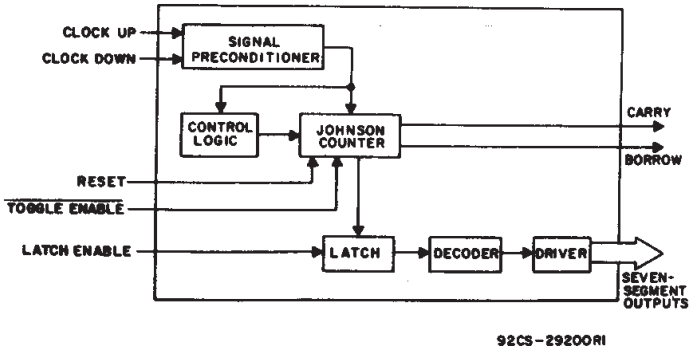


Fig. 1 - Functional diagram.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY-VOLTAGE RANGE, (V_{DD})
Voltages referenced to V_{SS} Terminal) ..... -0.5V to +20V
INPUT VOLTAGE RANGE, ALL INPUTS ..... -0.5V to V_{DD} +0.5V
DC INPUT CURRENT, ANY ONE INPUT ..... ±10mA
POWER DISSIPATION PER PACKAGE (P_D):
For T_A = -55°C to +100°C 500mW
For T_A = +100°C to +125°C ..... Derate Linearity at 12mW/°C to 200mW
DEVICE DISSIPATION PER OUTPUT TRANSISTOR
FOR T_A = FULL PACKAGE-TEMPERATURE RANGE (All Package Types) 100mW
OPERATING-TEMPERATURE RANGE (T_A) ..... -55°C to +125°C
STORAGE TEMPERATURE RANGE (T_{stg}) ..... -65°C to +150°C
LEAD TEMPERATURE (DURING SOLDERING):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10s max ..... +265°C

RECOMMENDED OPERATING CONDITIONS

For maximum reliability, nominal operating conditions should be selected so that operation is always within the following ranges:

CHARACTERISTIC	V_{DD} V	LIMITS MIN.	LIMITS MAX.	UNITS
Supply-Voltage Range (For T_A = Full Package Temperature Range)	—	3	18	V
Clock Input Frequency (f_{CL}) (Sum of CL_{UP} & CL_{DN} Freqs.)	5	—	1	MHz
	10	—	3	
	15	—	5	
Clock Pulse Width (t_W)	5	110	—	ns
	10	40	—	
	15	30	—	
Latch Enable Pulse Width	5	110	—	
	10	30	—	
	15	24	—	
Reset Removal-Time	5	550	—	
	10	200	—	
	15	130	—	
Reset Pulse Width	5	350	—	
	10	170	—	
	15	120	—	

STATIC ELECTRICAL CHARACTERISTICS

Characteristic	Conditions				LIMITS AT INDICATED TEMPERATURES (°C)							Units
	I_{OH} (mA)	V_{OH} (V)	V_{IN} (V)	V_{DD} (V)	-55	-40	+85	+125	+25 Min.	+25 Typ.	+25 Max.	
Quiescent Device Current Max. I_{DD}	—	—	—	5	5	5	150	150	—	0.04	5	μA
	—	—	—	10	10	10	300	300	—	0.04	10	
	—	—	—	15	20	20	600	600	—	0.04	20	
	—	—	—	20	100	100	3000	3000	—	0.08	100	
Output Voltage Low-Level Max. V_{OL}	—	—	0,5	5	0.05				—	0	0.05	V
	—	—	0,10	10	0.05				—	0	0.05	
	—	—	0,15	15	0.05				—	0	0.05	
High-Level Min. V_{OH}	—	—	0,5	5	—	—	—	—	—	4.55	—	V
	—	—	0,10	10	—	—	—	—	—	9.55	—	
	—	—	0,15	15	—	—	—	—	—	14.55	—	
Input Low Voltage Max. V_{IL}	—	0.5, 3.8	—	5	1.5				—	—	1.5	V
	—	1, 8.8	—	10	3				—	—	3	
	—	1.5, 13.8	—	15	4				—	—	4	
Input High Voltage Min. V_{IH}	—	0.5, 3.8	—	5	3.8				3.5	—	—	V
	—	1, 8.8	—	10	7				7	—	—	
	—	1.5, 13.8	—	15	11				11	—	—	
7-Segment Outputs Output Drive Voltage, High Min. V_{OH}	■	—	—	5	3.9		4		3.9	4.5	—	V
	-5	—	—		3.65		3.7		3.7	4.3	—	
	-10	—	—		3.55		3.65		3.65	4.25	—	
	-15	—	—		3.5		3.5		3.6	4.15	—	
	-20	—	—		3.45		3.35		3.45	4	—	
	-25	—	—		3.4		3.3		3.4	3.9	—	
	■	—	—	10	8.75		8.85		8.75	9.5	—	
	-5	—	—		8.45		8.55		8.55	9.3	—	
	-10	—	—		8.42		8.5		8.5	9.25	—	
	-15	—	—		8.4		8.47		8.47	9.2	—	
	-20	—	—		8.4		8.40		8.45	9.1	—	
	-25	—	—		8.3		8.25		8.3	9	—	
	■	—	—	15	13.8		13.9		13.8	14.5	—	
	-5	—	—		13.65		13.75		13.75	14.35	—	
	-10	—	—		13.6		13.72		13.72	14.3	—	
	-15	—	—		13.6		13.7		13.7	14.2	—	
	-20	—	—		13.6		13.6		13.65	14.1	—	
	-25	—	—		13.3		13.25		13.3	14.0	—	
7-Segment Outputs Output Low (Sink) Current Min. I_{OL}	—	0.4	0, 5	5	1.28	1.22	0.84	0.72	1	2	—	mA
	—	0.5	0, 10	10	3.2	3	2.2	1.8	2.6	5.2	—	
	—	1.5	0, 15	15	8.4	8	5.6	4.8	6.8	13.6	—	
Carry Outputs Output Low (Sink) Current Min. I_{OL}	—	0.4	0, 5	5	0.64	0.61	0.42	0.36	0.51	1	—	
	—	0.5	0, 10	10	1.6	1.5	1.1	0.9	1.3	2.6	—	
	—	1.5	0, 15	15	4.2	4	2.8	2.4	3.4	6.8	—	
Output High (Source) Current Min. I_{OH}	—	4.6	0, 5	5	-0.64	-0.61	-0.42	-0.36	-0.51	-1	—	
	—	2.5	0, 5	5	-2	-1.8	-1.3	-1.15	-1.6	-3.2	—	
	—	9.5	0, 10	10	-1.6	-1.5	-1.1	-0.9	-1.3	-2.6	—	
	—	13.5	0, 15	15	-4.2	-4	-2.8	-2.4	-3.4	-6.8	—	
Input Current Max. I_{IN}	—	0, 18	0, 18	18	±0.1	±0.1	±1	±1	—	$\pm 10^{-5}$	±0.1	μA

■ 0(10 μA)

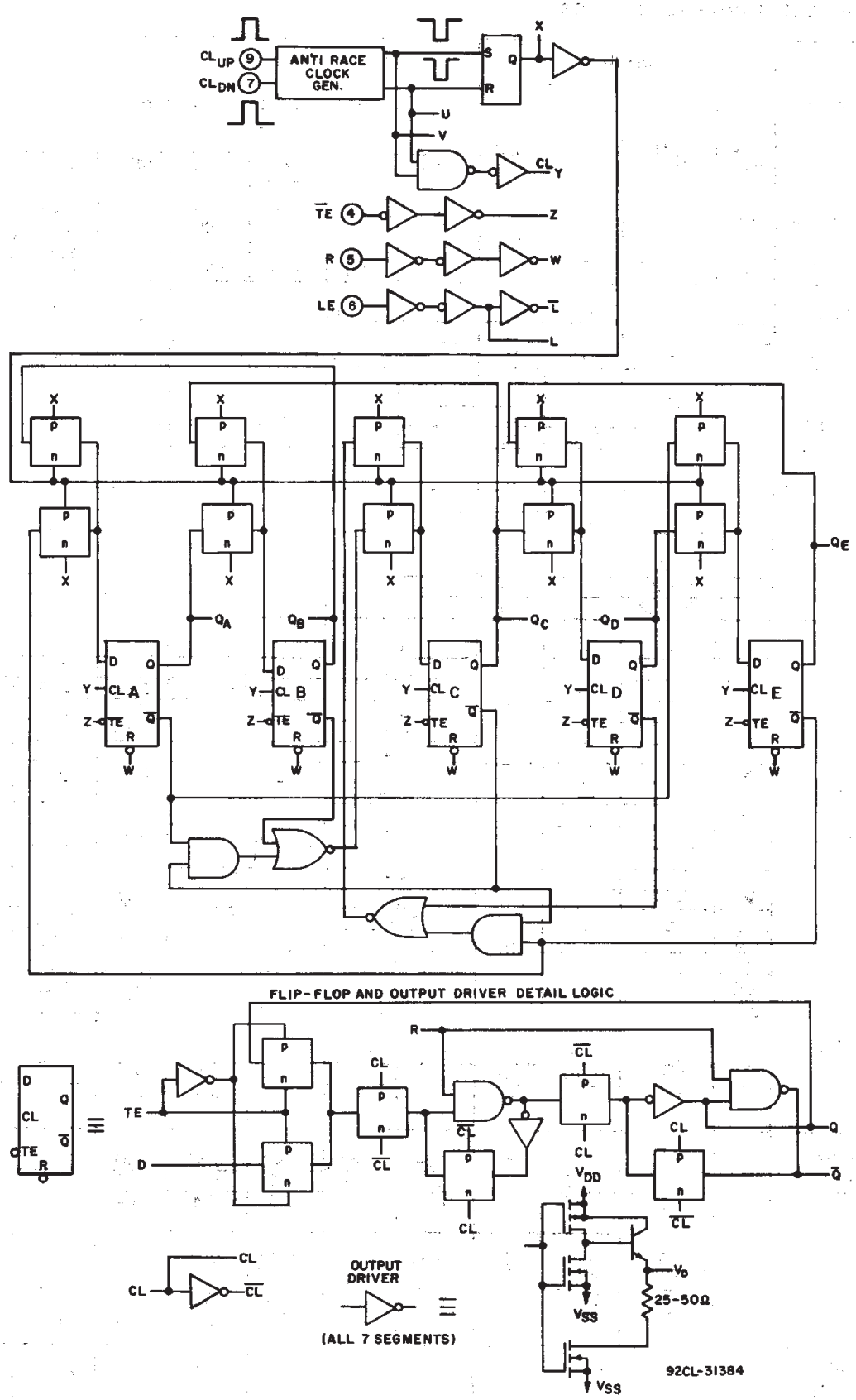


Fig. 2 - Logic diagram with flip-flop and output-driver details.
(cont'd on page 5)

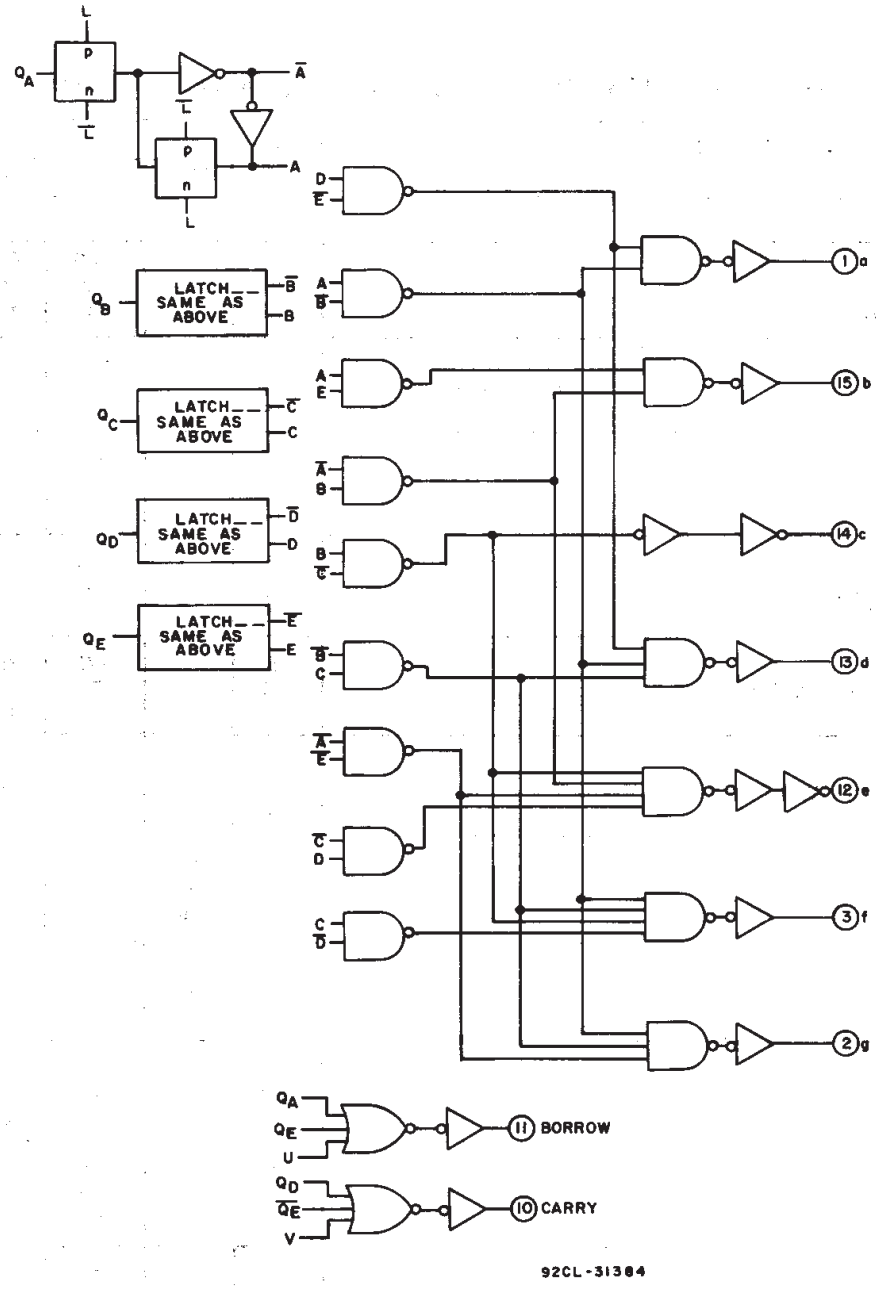
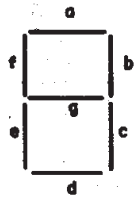
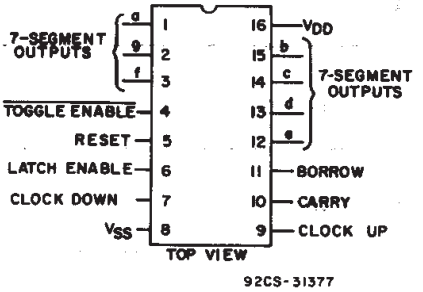


Fig. 2 - Logic diagram with flip-flop and output-driver details.

DISPLAY SEGMENTS



TERMINAL ASSIGNMENT



DYNAMIC ELECTRICAL CHARACTERISTICS at T_A = 25° C, Input t_r, t_f = 20 ns, C_L = 50 pF, R_L = 200 kΩ

CHARACTERISTIC		V_{DD} (V)	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
Clock Up/Clock Down						
Propagation Delay Time: Clock to Carry or Borrow	t_{PLH}, t_{PHL}	5	—	300	600	ns
		10	—	100	200	
		15	—	70	140	
Clock to Segment	t_{PLH}, t_{PHL}	5	—	925	1850	
		10	—	360	720	
		15	—	250	500	
Minimum Clock Pulse Width		5	—	55	110	
		10	—	20	40	
		15	—	15	30	
Maximum Clock Input Frequency (Sum of CL_{UP} & CL_{DN} F)	f_{CL}	5	1	2.5	—	MHz
		10	3	6	—	
		15	5	8.5	—	
Minimum Toggle Enable Pulse Width		5	—	175	350	ns
		10	—	75	150	
		15	—	55	110	
Minimum Latch Enable Pulse Width		5	—	55	110	
		10	—	15	30	
		15	—	12	24	
Output Pulse Width: Carry		5	115	230	—	
		10	60	120	—	
		15	40	75	—	
Borrow		5	140	275	—	
		10	65	130	—	
		15	45	85	—	
Transition Time: Carry or Borrow	t_{TLH}, t_{THL}	5	—	85	170	
		10	—	45	90	
		15	—	30	60	
Minimum Delay Time Between CL_{UP} & CL_{DN}		5	—	100	—	
		10	—	80	—	
		15	—	60	—	
Maximum Clock Rise or Fall Time	t_rCL, t_fCL	5	—	—	15	μs
		10	—	—	15	
		15	—	—	15	
Reset						
Propagation Delay Time Reset to Output	t_{PLH}, t_{PHL}	5	—	650	1300	ns
		10	—	350	700	
		15	—	160	320	
Minimum Reset Removal Time		5	—	-275	0	
		10	—	-100	0	
		15	—	-65	0	
Minimum Reset Pulse Width		5	—	175	350	
		10	—	85	170	
		15	—	60	120	

TRUTH TABLE

CLOCK UP *	CLOCK DOWN *	LATCH ENABLE	TOGGLE ENABLE	RESET	COUNTER	DISPLAY
⎍	X	0	0	0	Increments by 1	Follows Counter
X	⎍	0	0	0	Decrements by 1	Follows Counter
⎎	⎎	X	X	0	No Change	No Change
X	X	1	X	1	Goes to 00000	Remains Fixed
X	X	0	X	1	Goes to 00000	Follows Counter (Display = 0)
X	X	X	1	0	Inhibited	Remains Fixed
⎍	X	1	0	0	Increments by 1	Remains Fixed
X	⎍	1	0	0	Decrements by 1	Remains Fixed

X = Don't Care 1 = High State 0 = Low State

* Typically 100 ns between clock-up and clock-down positive transitions are required to ensure proper counting.

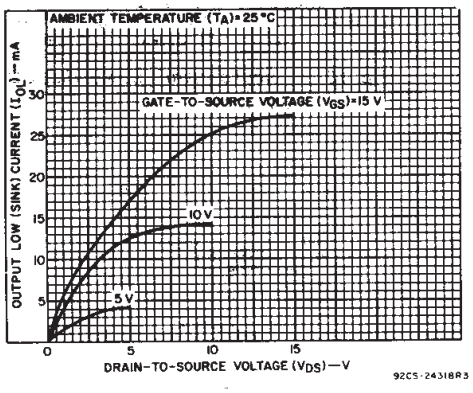


Fig. 3 - Typical carry or borrow output low (sink) current characteristics.

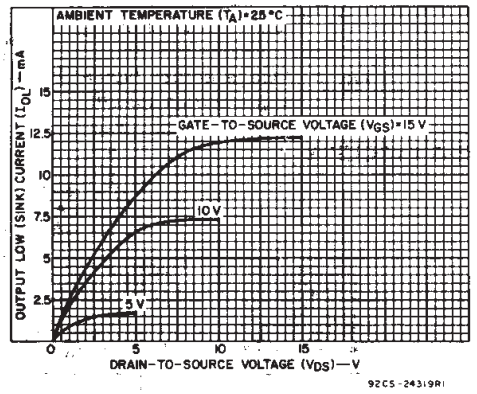


Fig. 4 - Minimum carry or borrow output low (sink) current characteristics.

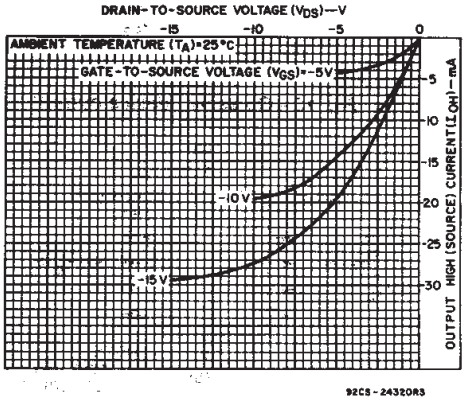


Fig. 5 - Typical carry or borrow output high (source) current characteristics.

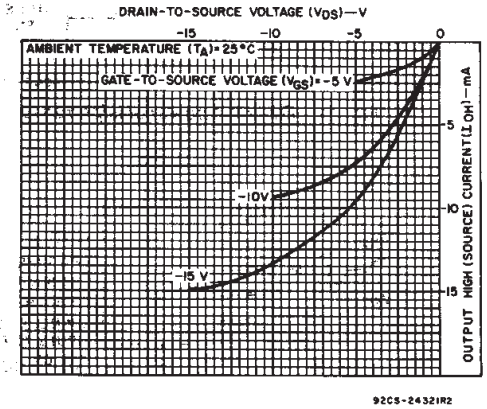


Fig. 6 - Minimum carry or borrow output high (source) current characteristics.

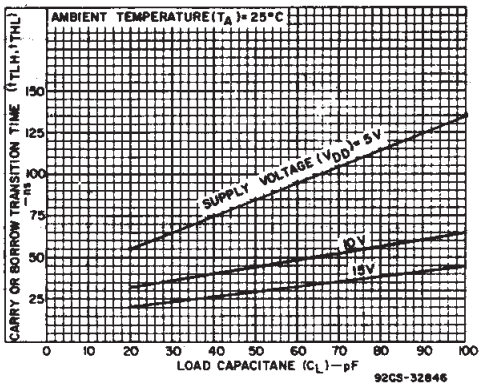


Fig. 7 - Typical carry or borrow transition time vs. load capacitance.

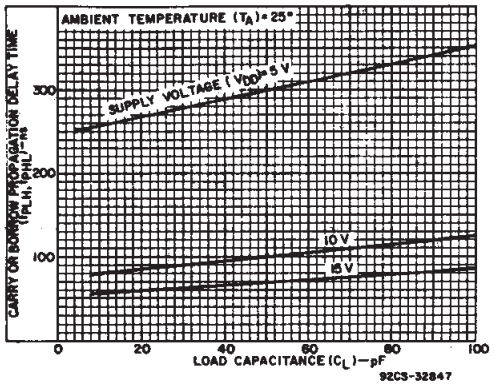


Fig. 8 - Typical carry or borrow propagation delay time vs. load capacitance.

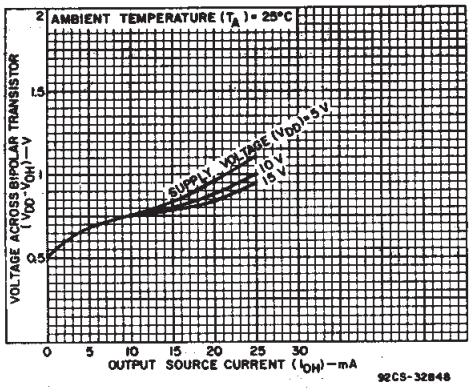


Fig. 9 - Voltage across bipolar transistor vs. output source current.

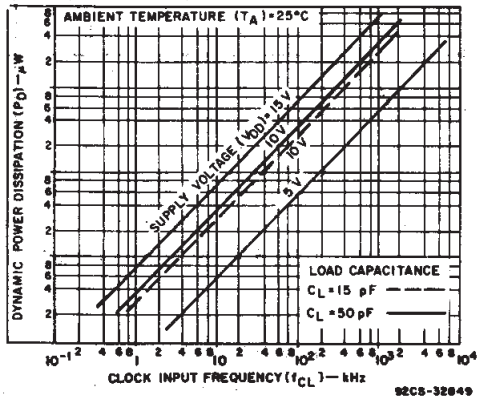


Fig. 10 - Typical dynamic power dissipation vs. frequency.

TEST CIRCUITS

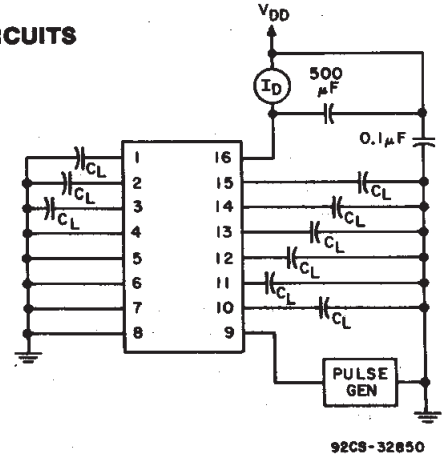


Fig. 11 - Dynamic power dissipation test circuit.

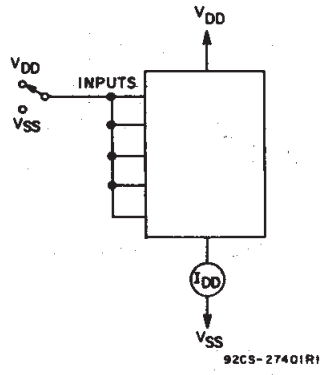


Fig. 12 - Quiescent device current.

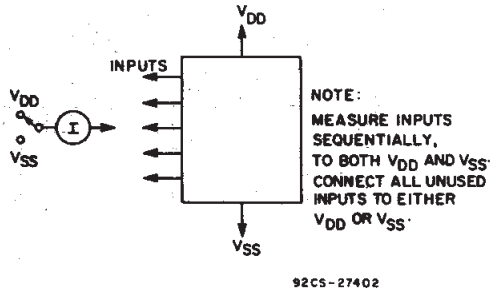


Fig. 13 - Input current.

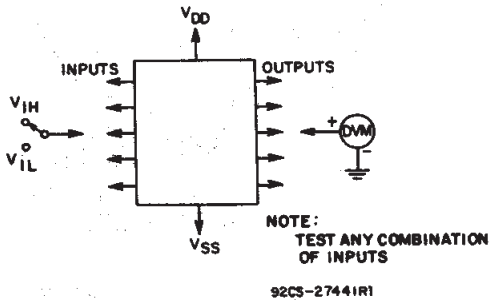


Fig. 14 - Input voltage.

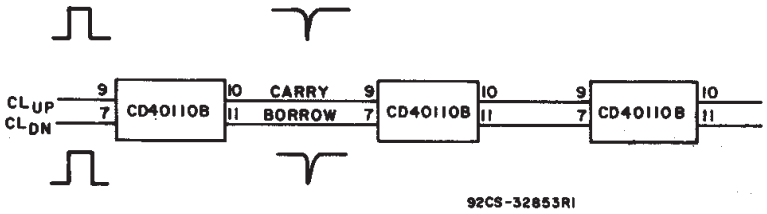
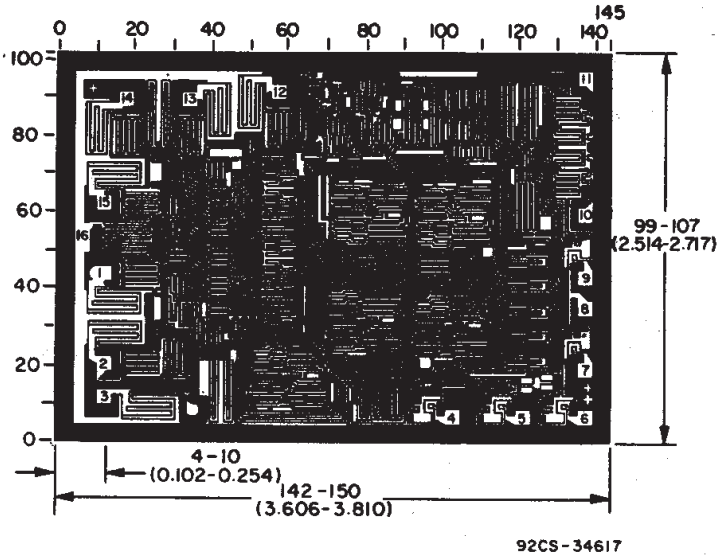


Fig. 15 - Cascading diagram.



Dimensions and pad layout for CD40110B.

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CD40110BE	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD40110BE
CD40110BE.A	Active	Production	PDIP (N) | 16	25 | TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD40110BE

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

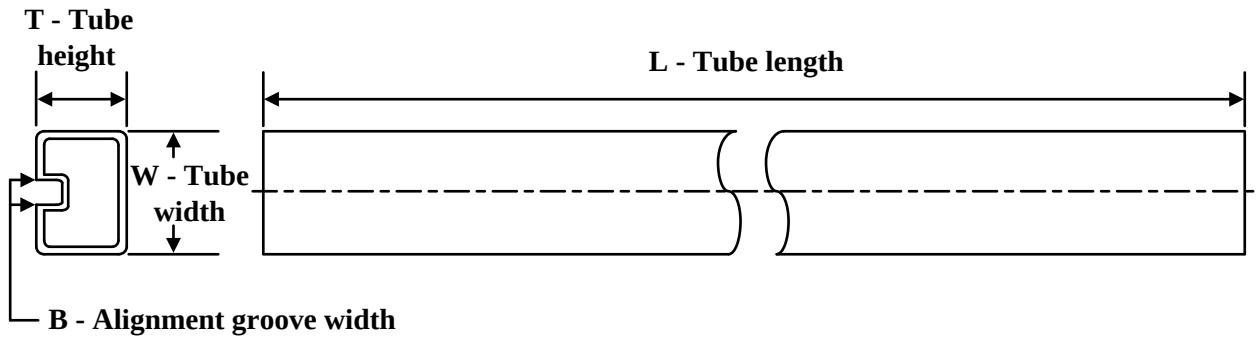
(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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TUBE



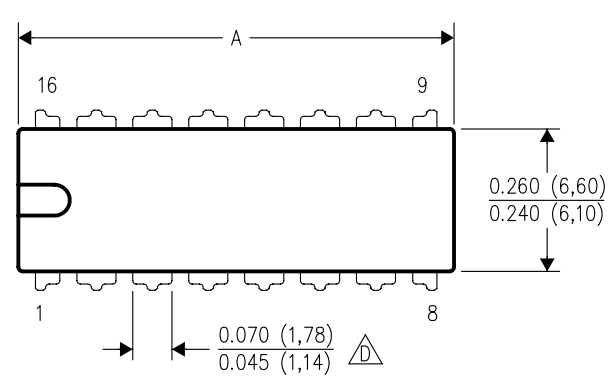
*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
CD40110BE	N	PDIP	16	25	506	13.97	11230	4.32
CD40110BE	N	PDIP	16	25	506	13.97	11230	4.32
CD40110BE.A	N	PDIP	16	25	506	13.97	11230	4.32
CD40110BE.A	N	PDIP	16	25	506	13.97	11230	4.32

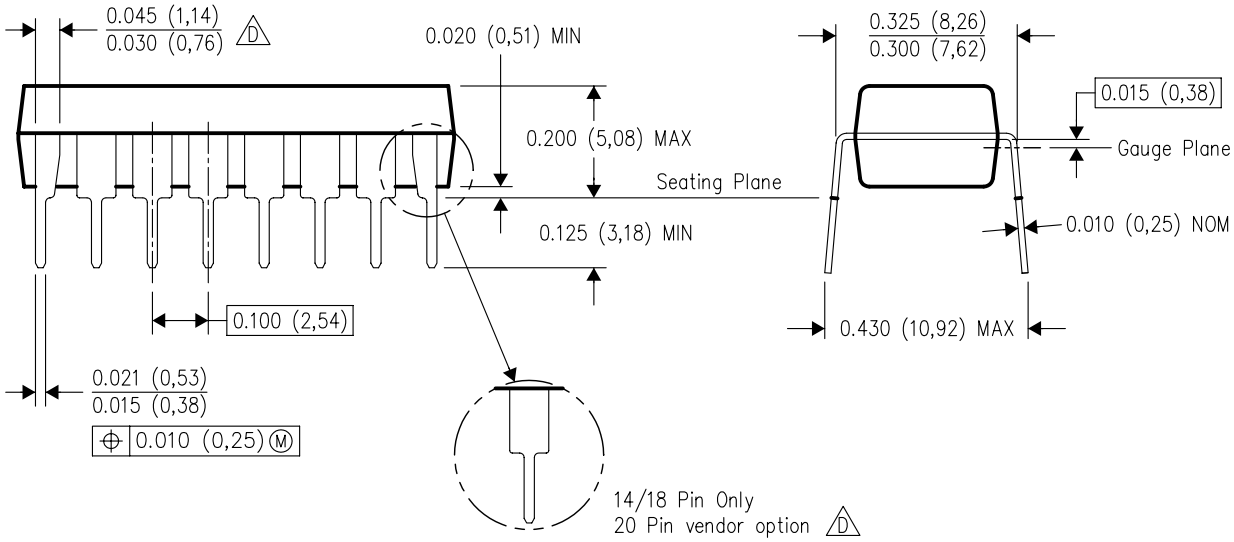
N (R–PDIP–T**)

PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN



DIM \ PINS **	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS–001 VARIATION (C)	AA	BB	AC	AD



4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

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Last updated 10/2025